

Data Mining: Data

Lecture Notes for Chapter 2

Introduction to Data Mining , 2nd Edition

by

Tan, Steinbach, Kumar

Outline

- Attributes and Objects
- Types of Data
- Data Quality
- Similarity and Distance
- Data Preprocessing

What is Data?

- Collection of **data objects** and their **attributes**
- An **attribute** is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, dimension, or feature
- A collection of attributes describe an **object**
 - Object is also known as record, point, case, sample, entity, or instance

Attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Objects

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Attribute Values

- **Attribute values** are numbers or symbols assigned to an attribute for a particular object
- Distinction between attributes and attribute values
 - Same attribute can be mapped to different attribute values *for a given sample*
 - ◆ Example: height can be measured in feet or meters
 - Different attributes can be mapped to the same *range* set of values
 - ◆ Example: Attribute values for ID and age are integers
 - But properties of an attribute can be different than the properties of the values used to represent the attribute

Clearly distinguish the attribute from its value.

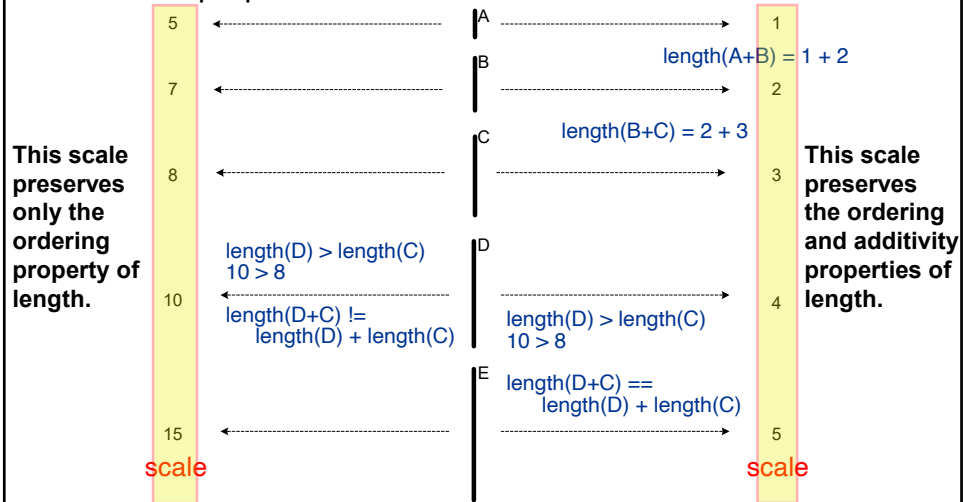
Attribute: Zip code
value: 32309

Although the value is an integer, the attribute is categorical. It's the attribute's properties — not the form of the value — that determine which transformations and analyses are appropriate

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Measurement of Length

- The way you measure an attribute may not match the attributes properties.



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Types of Attributes

- There are different types of attributes
 - **Nominal**
 - ◆ Examples: ID numbers, eye color, zip codes
 - **Ordinal** (numerical values)
 - ◆ Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}
 - **Interval** (can subtract values)
 - ◆ Examples: calendar dates, temperatures in Celsius or Fahrenheit.
 - **Ratio** (There is a zero, which allows ratio of two values) (zero denotes "an absence of")
 - ◆ Examples: temperature in Kelvin, length, counts, elapsed time (e.g., time to run a race)

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Properties of Attribute Values

- The type of an attribute depends on which of the following properties/operations it possesses:

- Distinctness: $= \neq$
- Order: $< >$
- Differences are meaningful : $+ -$
- Ratios are meaningful $* /$
- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & meaningful differences
- Ratio attribute: all 4 properties/operations

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Difference Between Ratio and Interval

- Is it physically meaningful to say that a temperature of 10° is twice that of 5° on

- the Celsius scale? No
- the Fahrenheit scale? No
- the Kelvin scale? Yes

Zero Kelvin is an
absence of temperature

- Consider measuring the height above average
 - If Bill's height is three inches above average and Bob's height is six inches above average, then would we say that Bob is twice as tall as Bill?
 - Is this situation analogous to that of temperature?

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Categorical Qualitative	Attribute Type	Description	Examples	Operations
	Nominal	Nominal attribute values only distinguish. ($=$, $\neq$)	zip codes, employee ID numbers, eye color, sex: { <i>male</i> , <i>female</i> }	mode, entropy, contingency correlation, χ^2 test <i>probabilities</i>
	Ordinal	Ordinal attribute values also order objects. ($<$, $>$)	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
	Interval	For interval attributes, differences between values are meaningful. ($+$, $-$)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Numeric Quantitative	Ratio	For ratio variables, both differences and ratios are meaningful. ($*$, $/$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current	geometric mean, harmonic mean, percent variation

Sometimes the answer is not clear: i.e., distance from the origin (positive or negative). Depends on the origin.

This categorization of attributes is due to S. S. Stevens

Why entropy? Because it is based on probabilities, in this case, frequencies

Ratio attributes are always positive

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Categorical Qualitative	Attribute Type	Transformation	Comments
	Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
	Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function (in addition to Nominal transformations)	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}.
	Interval	$new_value = a * old_value + b$ where a and b are constants (in addition to Ordinal transformations)	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Numeric Quantitative	Ratio	$new_value = a * old_value$ where 'a' is positive (in addition to Interval transformations)	Length can be measured in meters or feet.

This categorization of attributes is due to S. S. Stevens

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Discrete and Continuous Attributes

□ Discrete Attribute

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables.
- Note: **binary attributes** are a special case of discrete attributes

□ Continuous Attribute

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits. That makes them discrete (technically), but not in practice.
- Continuous attributes are typically represented as floating-point variables.

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Asymmetric Attributes

□ Only presence (a non-zero attribute value) is regarded as important

- ◆ Words present in documents
- ◆ Items present in customer transactions

□ If we met a friend in the grocery store would we ever say the following?

“I see our purchases are very similar since we didn’t buy most of the same things.”

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Critiques of the attribute categorization

- Incomplete There are additional categories, for example:
 - Asymmetric binary
 - Cyclical
 - Multivariate
 - Partially ordered (family tree)
 - Partial membership (e.g. 0.8 membership in the set “tall”)
 - Relationships between the data
- Real data is approximate and noisy
 - This can complicate recognition of the proper attribute type
 - Treating one attribute type as another may be approximately correct

Multimedia, temporal, textual, continuous, categorical, discrete, spatial, sequential, graph-based, and more.

Read more: <https://chat.openai.com/share/3af6b337-a4bb-4108-becf-f7133f645a2c>

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Key Messages for Attribute Types

- The types of operations you choose should be “meaningful” for the type of data you have
 - Distinctness, order, meaningful intervals, and meaningful ratios are only four (among many possible) properties of data
 - The data type you see – often numbers or strings – may not capture all the properties or may suggest properties that are not present
 - Analysis may depend on these other properties of the data
 - ◆ Many statistical analyses depend only on the distribution
 - In the end, what is meaningful can be specific to domain

Additional properties

- granularity
- dimensionality
- quality
- relevance
- variability
- spatio-temporal
- scalability
- heterogeneity
- noise and outliers
- balance and imbalance
- sparsity
- resolution
- size
- distribution

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Important Characteristics of Data

- Dimensionality (number of attributes)
 - ◆ High dimensional data brings a number of challenges
- Sparsity
 - ◆ Only presence counts
- Resolution
 - ◆ Patterns depend on the scale
- Size
 - ◆ Type of analysis may depend on size of data

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Types of data sets

- Record (rows of attributes)
 - Data Matrix
 - Document Data
 - Transaction Data
- Graph (nodes and edges)
 - World Wide Web
 - Molecular Structures
- Ordered
 - Spatial Data
 - Temporal Data
 - Sequential Data
 - Genetic Sequence Data

There are more types:
e.g., cyclic

A dataset can be of
multiple types

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Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Usually a single target. There could be several targets. Several targets can be interpreted as a single target in a higher dimensional space.

e.g., 3 columns of weights could be interpreted as a single column of points in 3D space.

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Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such a data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Or a row could correspond to multiple attributes, each in a higher-dimensional space.

Different interpretations of the same data lead to different algorithms. The number of possible algorithms is nearly infinite.

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Document Data

- Each document becomes a 'term' vector
 - Each term is a component (attribute) of the vector
 - The value of each component is the **number of times** the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

It could also be whether the word appears or not.

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Transaction Data

- A special type of data, where
 - Each transaction involves a set of items.
 - For example, consider a grocery store. **The set of products purchased by a customer during one shopping trip** constitute a **transaction**, while **the individual products** that were purchased are the **items**.
 - Can represent transaction data as record data

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Each row is a transaction

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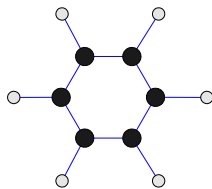
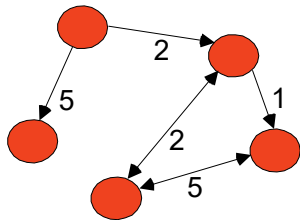
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Alternative representation:

	Milk	Coke	Beer	Bread	Diaper
1	1	1		1	
2			1	1	
3	1	1	1		1
4	1		1	1	1
5	1	1			1

Graph Data

- Examples: Generic graph, a molecule, and webpages



Benzene Molecule: C₆H₆

Useful Links:

- [Bibliography](#)
- Other Useful Web sites
 - [ACM SIGKDD](#)
 - [KDnuggets](#)
 - [The Data Mine](#)

Knowledge Discovery and Data Mining Bibliography

(Gets updated frequently, so visit often!)

- [Books](#)
- [General Data Mining](#)

Book References in Data Mining and Knowledge Discovery

Usama Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy Iyerasamy, "Advances in Knowledge Discovery and Data Mining", AAAI Press/the MIT Press, 1996.

J. Ross Quinlan, "C4.5: Programs for Machine Learning", Morgan Kaufmann Publishers, 1993.

Michael Berry and Gordon Linoff, "Data Mining Techniques (For Marketing, Sales, and Customer Support)", John Wiley & Sons, 1997.

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Christopher Matheus, Philip Chan, and Gregory Piatetsky-Shapiro, "Systems for knowledge Discovery in databases", IEEE Transactions on Knowledge and Data Engineering, 5(6):903-913, December 1993.

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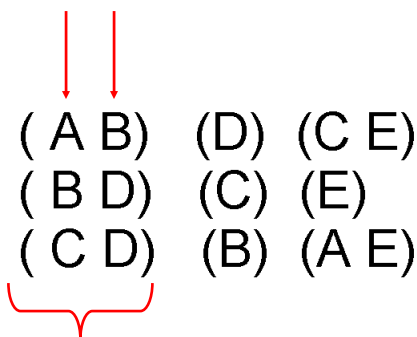
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Ordered Data

- Sequences of transactions (with or without timestamps)

Items/Events



An element of
the sequence

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Ordered Data

□ Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCCGCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

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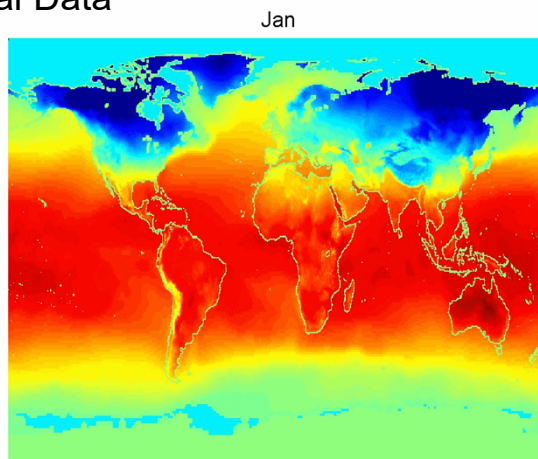
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Ordered Data

□ Spatio-Temporal Data

**Average Monthly
Temperature of
land and ocean**



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Data Quality

- Poor data quality negatively affects many data processing efforts

- Data mining example: a classification model for detecting people who are loan risks is built using poor data
 - Some credit-worthy candidates are denied loans
 - More loans are given to individuals that default

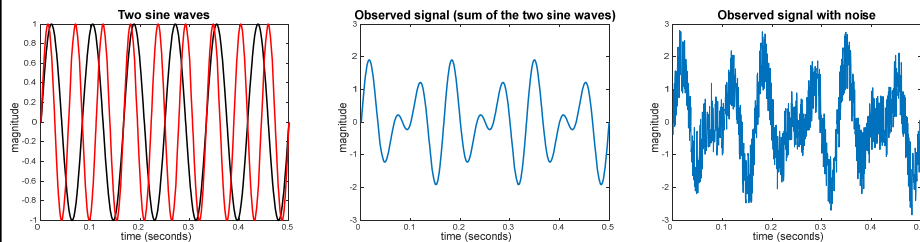
Data Quality ...

- What kinds of data quality problems?
- How can we detect problems with the data?
- What can we do about these problems?

- Examples of data quality problems:
 - Noise and outliers
 - Wrong data
 - Fake data
 - Missing values
 - Duplicate data

Noise

- For objects, noise is an extraneous object
- For attributes, noise refers to modification of original values
 - Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen
 - The figures below show two sine waves of the same magnitude and different frequencies, the waves combined, and the two sine waves with random noise
 - ◆ The magnitude and shape of the original signal is distorted



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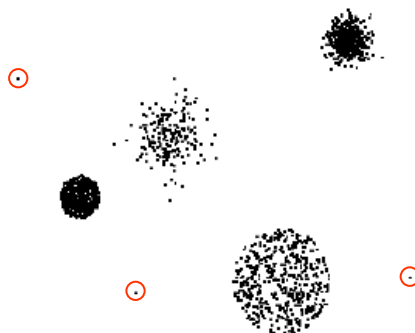
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Outliers

- **Outliers** are data objects with characteristics that are considerably different than most of the other data objects in the data set
 - **Case 1:** Outliers are noise that interferes with data analysis
 - **Case 2:** Outliers are the goal of our analysis
 - ◆ Credit card fraud
 - ◆ Intrusion detection

Outliers not due to noise usually have a reason that can be identified (i.e., power outages, which occur rarely).

□ **Causes?**



Noise is not necessary for outliers.
e.g. the length of time a light bulb shines before burning out. If the burnout occurs after 24 hours, that is an outlier.

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Missing Values

- Reasons for missing values
 - Information is not collected (e.g., people decline to give their age and weight)
 - Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)
- Handling missing values
 - Eliminate data objects or variables
 - Estimate missing values
 - ◆ Example: time series of temperature
 - ◆ Example: census results
 - Ignore the missing value during analysis

What if 90% of the rows have missing values?

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Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
 - Major issue when merging data from heterogeneous sources
- Examples:
 - Same person with multiple email addresses
- Data cleaning
 - Process of dealing with duplicate data issues
- When should duplicate data not be removed?

A dataset has 100 attributes. Two rows have 90 attributes that are identical. What to do?

What if only 2 attributes are identical?

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